Aspen Johnson

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Education

Valdosta State University
B.S in Data Science: Computational Sci. & Eng. Track, Minors in Physics and Math — *Expected: 2028, GPA: N/A*

Palm Beach State College | Academic transition year
A.A in STEM: Mechanical/ Aerospace Engineering Track — *Aug 2025 - May 2026, GPA: 3.2*

Mercer University
Coursework in Finance, Mathematics, and Economics — *June 2022 - May 2025, GPA: 3.0*

Research Experience

Undergraduate Research | Long-Memory Time Series & Seismicity Modelling — June 2026 – October 2026
Faculty Guided Research w/ Prof. Szabolcs Blaszek — *Remote*

- Independently proposed and initiated a research investigation into earthquake dynamics with the aim of exploring the science behind seismic activity.
- Developed a Python-based data acquisition workflow using USGS earthquake catalog queries, compiling and analyzing over two million seismic observations spanning more than 50 years across California and Alaska.
- Investigated earthquake-count series, inter-event intervals, and seismic activity clustering as potential measures of temporal dependence, with an emphasis on **long-memory behavior** and **score-driven fractional integration models**.

Undergraduate Research | Computational Physics & Molecular Dynamics — April 2026 – Present
Faculty Guided Research w/ Prof. Shafat Mubin — *On-Site/Remote*

- Studied the **Velocity Autocorrelation Function (VACF)** to analyze how particle velocities evolve over time in molecular dynamics trajectories.
- Transitioning toward atomistic molecular dynamics research, studying LAMMPS simulation workflows, interatomic potentials, and numerical time integration in preparation for Spring 2027.

Undergraduate Research | Score-Driven Volatility Models — Jan 2025 – Dec 2025
Faculty Guided Research w/ Prof. Szabolcs Blaszek and Adam Kobor — *On-Site/Remote*

- Contributed to work showing that Hessian-driven gradient scaling can improve volatility-filter performance relative to existing scaling methods.
- Gained publication experience through faculty-guided research involving model interpretation, literature review, empirical analysis, and manuscript development.
- Analyzed financial time-series data across Bitcoin, S&P 500, Gold, U.S. REITs, and CHF/JPY from 2010–2025 using R, GAUSS, Excel, and Bloomberg Terminal data.

Publications & Research Presentations

An empirical application of improved gradient scaling for score-driven volatility filters — Published

- Selected as 1 of 15 research papers to present at the 2026 Atlanta Research Conference.
- Georgia Undergraduate Research Conference (GURC), November 2026 – Accepted for presentation.

Score-Driven Dynamic Fractional Integration with Heavy Tails: An Application to Earthquake Activity — Under Review

Projects

CourtSignal | *Assistive Technology for Hearing Impaired Athletes* — In Progress
Independent Research & Engineering Project

- Conceived an assistive technology concept to improve real-time communication for Deaf and hard-of-hearing athletes through visual and haptic signaling.
- Investigating multimodal communication, human-centered design, and potential hardware-software integration to support athlete awareness, coach instruction, and team coordination during competition.
- Selected for the University System of Georgia and Georgia Tech's inaugural 2026 Startup Studio seminar, supporting customer discovery, entrepreneurship, and early-stage product validation.

Custom Cyberdeck — In Progress
Independent Engineering Project | Portable, Powerful, and Playable

- Learning the theory behind designing a compact, handheld computing system optimized for extended use during collegiate athletic travel, prioritizing ergonomics, portability, and battery efficiency.

Industry & Work Experience

Financial Analyst Intern May 2025 – Dec 2025
Zinzino USA *Remote*

- Reduced chargeback processing time by 87% through workflow digitization and documentation streamlining.
- Refined the charge back process within a few weeks of arrival, and curated a reference guide to aid future onboarding employees.
- Created clear documentation that enabled cross-departmental collaboration and made Finance operations easy for shareholders and upline leaders to understand.

Leadership, Athletics, and Service

Collegiate Women's Basketball Player June 2022 – Present
Mercer University | Valdosta State University *NCAA Division I | Division II*

- Competed in NCAA Division I and II women's basketball, balancing 20–25+ hours of weekly athletic commitments with undergraduate research and academic coursework.
- Applied firsthand experience in athletic communication, team coordination, and rapid decision-making to the development of accessibility-focused sports technology.

Leadership, Community Engagement & Extracurriculars

Read2Succeed | *Tutoring elementary students to aid in building literary in Macon, Georgia*
Putnam Mathematics Competition Prep | *Independent Study*
Youth Sports Volunteer | *Assisted young athletes K-12 in a variety of sports.*
Weekly Physics Study Group Member | *Going through* ***Helliwell's Modern Classical Mechanics*** *text*
Book Club Fellow | *Selected as one of ten students to participate through Mercer Universities Economics Department*
Content Creation | *Live streamed my workflow on YouTube amassing ∼20,000 views*
AWS Campus Prep Series Participant | *Completed a Summer-long series of webinars regarding Amazon Careers and Cloud Services*

Technical Skills

Languages in order of proficiency: Python, Java, LaTeX, HTML/CSS, R, Gauss, C/C++, JavaScript, Octave, TypeScript, JavaScript
Developer Tools: Git, CLI, VSCode

Other

Clubs/Organizations:NSBE, SWE, and Codecademy's Women in Tech
Relevant Coursework: Calculus I–III, Linear Algebra, Differential Equations (ODE), Probability and Statistics, Data Structures, Programming, Intro Econometrics, Time Series Analysis in R, Physics I